

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) constitutes a promising framework for improving the reasoning capabilities of large language models. Existing instantiations such as GRPO, however, frequently exhibit vanishing policy gradients. We attribute this failure mode to two coupled mechanisms: the use of a fixed clipping bound on token-level probability ratios, and the standardization of rewards within a group in which the realized rewards are identical. Jointly, these mechanisms yield uninformative gradient updates and low utilization of the sampled responses. To address both mechanisms, we introduce **Dynamic Clipping Policy Optimization (DCPO)**, which comprises (i) a dynamic clipping scheme that adapts the per-token clipping bounds as a function of the token-specific prior probability under the old policy, thereby enlarging the admissible update region for token-level exploration, and (ii) a smooth advantage standardization procedure that normalizes rewards over cumulative training steps, thereby increasing the response-level utilization of the generated responses. DCPO attains state-of-the-art performance across four benchmarks and four distinct base models. Concretely, on the AIME24 benchmark with Qwen2.5-Math-7B, DCPO achieves an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-sample decoding, exceeding DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On the AIME25 benchmark with Qwen2.5-14B, DCPO attains (23.3/19.0), exceeding GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). In aggregate over the four models, DCPO raises the nonzero-advantage ratio by an average of 28 percentage points relative to GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude relative to both GRPO and DAPO, while delivering superior task performance. These results establish that DCPO more efficiently exploits generated data for reinforcement learning of large language models.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central research focus (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) is a promising technique for eliciting such capabilities: it optimizes an LLM policy against rule-based outcome rewards, and has been shown to improve performance in domains that demand multi-step reflective decision-making, notably mathematics and coding (DeepSeek-AI et al., 2024).

RLVR is commonly instantiated through the GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of policy-gradient algorithms. While the empirical robustness of GRPO is well documented, subsequent analyses—most notably DAPO (Yu et al., 2025)—have identified two structural limitations. First, GRPO is prone to entropy collapse, in which the estimated advantages contract toward zero in the later phases of training and the policy distribution loses stochasticity. Second, the clipping operator, when misconfigured, suppresses learning signals carried by infrequently sampled tokens, thereby narrowing the effective exploration support of the policy.

DAPO addresses these limitations through two mechanisms: a Clip-Higher rule that relaxes the upper clipping bound to counteract entropy collapse, and a Dynamic Sampling rule that discards prompts whose sampled rewards are uniform, guaranteeing that every optimization batch contains only responses with nonzero gradients. These mechanisms, however, introduce two limitations of their own. (i) **Response-level inefficiency**: dynamic sampling discards a substantial fraction of generated

responses, reducing sampling efficiency and slowing convergence. (ii) **Token-level clipping deficit**: our analysis indicates that the variance-optimal clipping bound is token-dependent, so a single upper bound shared across all tokens is provably suboptimal. Although alternative methods avoid explicitly discarding zero-gradient responses, both limitations persist across this family of algorithms.

In this work, we propose **Dynamic Clipping Policy Optimization (DCPO)**, which targets the two limitations above through three coordinated components. First, we introduce a **smooth advantage standardization that normalizes advantages over cumulative training steps**, which remedies the vanishing-gradient problem induced by zeroing the advantage whenever a group’s rewards are identical. The procedure fuses the reward statistics of all previously generated responses to a prompt with those of the responses generated at the current step, thereby recovering informative response-level signal (see Section 2). Second, we propose a **dynamic clipping scheme that sets the per-token clipping bounds as an explicit function of the token-specific probability under the old policy**, in contrast to prescribing fixed, hand-tuned upper and lower bounds. Concretely, tokens assigned lower prior probability receive wider clipping bounds, enlarging the admissible update region available for token-level exploration (see Section 2 and Appendix A.6). Third, DCPO computes the loss as a mean over the tokens within each individual response rather than over all tokens in a batch; this normalization preserves the relative advantage ordering among responses to a common prompt and eliminates the dilution incurred by batch-level averaging.

For evaluation, we construct a training corpus by merging Math-DAPO-17k (Yu et al., 2025) with levels 3–5 of the MATH dataset (Liu et al., 2025), and assess performance on four standard mathematical-reasoning benchmarks. We run all experiments across four base models, using GRPO, DAPO, and—where available—GSPO as baselines. DCPO consistently attains the strongest results, with the largest margins on the challenging AIME benchmarks: under 32-sample decoding, DCPO-7B scores 38.8 on AIME24-Avg@32 versus 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), while DCPO-14B reaches 19.0 on AIME25-Avg@32 versus 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). In addition, DCPO increases the average nonzero-advantage ratio by 28 percentage points over GRPO, doubles training efficiency over DAPO, and reduces the token clipping ratio by an order of magnitude relative to all baselines, while delivering superior task performance.

A formal account of the loss and advantage-estimation procedures of the prior algorithms GRPO, DAPO, and GSPO is provided in Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a standard mechanism in reinforcement learning for stabilizing policy updates and bounding the magnitude of parameter displacement per optimization step. Its role is most critical under importance sampling in the off-policy regime. Reweighting samples by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$ yields an unbiased estimator of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$; however, the variance of this estimator can be substantially inflated relative to the on-policy variance, as quantified in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right)^2 \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right)^2 p(x) dx \quad (1)$$

To control this variance inflation, prior work (e.g., Schulman et al. (2017)) constrains the probability ratio to a fixed symmetric interval parameterized by the hyperparameter ϵ , as stated in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping interval, however, restricts the policy’s exploration, particularly in regions where the old policy places low probability mass: there the admissible absolute change for a decreasing probability is inherently small, which curtails the magnitude of any meaningful policy update. Asymmetric bounds, as adopted in DAPO, only partially relax this constraint, and the underlying non-adaptivity remains. GSPO, in turn, discards an entire response whenever its sequence-level importance ratio falls outside the asymmetric interval, which forfeits informative tokens while retaining high-variance ones, and thereby destabilizes training.

From a variance–bias standpoint, a more principled formulation replaces the fixed interval with a probability-dependent constraint, formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the radicand by taking its maximum with zero yields a closed-form admissible interval for $r(x)$. The resulting **dynamic-adaptive clipping bounds** depend explicitly on $q(x)$ and on the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive interval widens as $q(x)$ decreases, thereby permitting greater policy exploration in low-probability regions. Following the dual-clipping construction of (Ye et al., 2020), we additionally impose a hard ceiling of 10 on the ratio for both positive and negative advantages, which prevents the widened interval from admitting excessively large importance weights and the attendant over-updating. The full derivation and analysis of these bounds within the importance-sampling framework are deferred to Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior methods such as GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing solely the rewards $R_{j=1,\dots,G}^i$ collected for the same prompt at step i , as given by Equation (11) in Appendix A.3. This step-local estimator has two failure modes. First, when sampling stochasticity produces identical rewards across the G responses at a step, the standardized advantage is identically zero, so the prompt contributes no gradient—despite potentially informative differences among the underlying reasoning trajectories. Second, under high-entropy sampling the empirical label counts can be highly skewed from step to step, inducing large fluctuations, and even sign reversals, in the standardized advantages, which destabilizes optimization.

Concurrently, Yue et al. (2025) report that RLVR chiefly reweights the likelihood of already-reachable correct samples rather than substantially displacing the base policy’s distribution. Under this premise, the reward distribution of responses to a given prompt can be modeled as stationary across training—i.e., drawn from a single global distribution—which motivates a *cumulative* standardization over all steps, defined in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the mean μ_{total}^i and standard deviation σ_{total}^i are accumulated over all responses generated for the prompt up to and including step i .

To damp the fluctuations of both the step-local estimator ($\hat{A}_{new,j}^i$) and the cumulative estimator ($\hat{A}_{total,j}^i$), we define two smoothed estimators, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each a convex combination of the two standardizations whose mixing weights are functions of the step index i , as given in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

As i grows, $\hat{S}A_{new,j}^i$ assigns increasing weight to the current-step distribution, biasing the estimate toward the latest policy outputs; conversely, $\hat{S}A_{total,j}^i$ assigns increasing weight to the cumulative distribution, treating the current step as a local correction within the broader distribution. The final advantage $\hat{A}_j^i$ is then taken as whichever of the two smoothed estimators has the smaller absolute value, a selection rule that suppresses the larger of the two fluctuation sources at each step and thereby

improves training stability, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt has entered optimization, every response generated for it at subsequent steps contributes to the update irrespective of whether its step-local advantage is zero. In particular, when a step yields identical rewards, its responses still update the policy with the residual advantage $\frac{1}{i} \hat{A}_{total,j}^i$, which retains otherwise-discarded learning signal and improves data efficiency.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)**—the objective Equation (13) of DAPO—equalizes the contribution of every token, but in doing so it distorts the relative weighting of responses of differing length. As a concrete illustration, consider two responses to the same prompt: response *A* with advantage 1 and length 500, and response *B* with advantage 0.5 and length 1500. TLM weights *A* by $\frac{500}{500+1500} = 0.25$ and *B* by $\frac{1500}{500+1500} = 0.75$, so their loss contributions are $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. Consequently *B* exerts the larger influence (0.375) despite its smaller advantage (0.5), inverting the intended ordering.

Conversely, the **Sequence-Level Mean loss (SLM)**—the objective Equation (12) of GRPO—averages over the batch and thereby attenuates each advantage by a factor of G , which for large batch sizes erodes the relative structure that standardization had established across responses. Because standardized advantages already encode the correct relative ordering across responses, we argue that averaging only over the $|o_i|$ tokens of a single response both preserves that ordering and distributes importance uniformly across the tokens of that response. We therefore average over the tokens within each response and omit the batch-level average over responses. We term this objective the **Only Token Mean loss (OTM)**, given in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

Following DAPO, the DCPO objective omits the KL-divergence penalty. The lower and upper clipping bounds ε_{low} and $\varepsilon_{\text{high}}$ are instantiated by the dynamic-adaptive bounds derived in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate on four base models spanning two families and a range of capacities: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (general-purpose base), Qwen2.5-Math-7B (math-specialized base), and Qwen2.5-14B (general-purpose base) (Yang et al., 2024a;b). Training uses a merged corpus of approximately 25k mathematical problems, formed by combining the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3–5 of the MATH subset (Hendrycks et al., 2021b; Liu et al., 2025). Every problem is rendered with the Qwen-Math template (see Section A.8) for both training and evaluation, ensuring a consistent prompt format across conditions.

Evaluation We measure performance on four widely used mathematical-reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, whose 500 problems make repeated sampling costly, we report Avg@1 only. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems), we report both Avg@1 and Avg@32 to obtain a statistically more reliable estimate under sampling.

- **Avg@1**: accuracy under greedy (argmax) decoding, i.e., the accuracy of the model’s single most probable prediction.
- **Avg@32**: accuracy averaged over 32 responses sampled per problem at temperature 1.0 and top_p 1.0; this estimator characterizes the robustness and stability of the learned policy distribution under stochastic decoding.

Baseline Settings We adopt GRPO and DAPO as baselines for all models, and additionally include GSPO for Qwen2.5-Math-7B and Qwen2.5-14B. The remaining configuration—including the per-method clipping thresholds ϵ_* , the codebase, and the shared training hyperparameters—is specified in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1: Benchmark performance across models and methods. Avg@1 denotes greedy decoding; Avg@32 denotes the mean over 32 sampled responses. Boldface marks the best value in each column; Δ reports the absolute improvement of DCPO over the indicated baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

Table 1 reports the performance of the four policy-optimization methods—GRPO, DAPO, GSPO, and the proposed DCPO—on the MATH500, AMC23, AIME24, and AIME25 benchmarks, under both greedy decoding (Avg@1) and sampling-based decoding (Avg@32), so that robustness and generalization are jointly characterized. Across every model scale, DCPO is superior or statistically comparable to GRPO, DAPO, and—where evaluated—GSPO, supporting its role as a general-purpose policy-optimization method for reinforcement learning of LLMs.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO attains the highest average scores (47.2/32.8), exceeding GRPO by (+1.2/+0.8) and DAPO by (+0.7/+0.4). On Qwen2.5-3B, DCPO yields (37.6/22.7), exceeding GRPO by (+1.3/+1.7) and remaining competitive with

DAPO—improving Avg@1 by (+1.0) while trailing on Avg@32 by only (-0.4). On Qwen2.5-Math-7B, DCPO reaches (57.1/45.2), exceeding GRPO by (+4.0/+3.6), DAPO by (+3.3/+2.8), and GSPO by (+1.9/+1.9). On the largest model tested, Qwen2.5-14B, DCPO attains (53.2/39.0), improving over GRPO by (+6.5/+7.7), over DAPO by (+1.3/+0.1), and over GSPO by (+4.2/+5.5) on Avg@1 and Avg@32, respectively. These results indicate that the improvements conferred by DCPO are consistent across model capacities and that its policy aligns more closely with the reward signal under both greedy and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) The most pronounced effect appears on AIME24 and AIME25, high-difficulty competition benchmarks that stress-test multi-step reasoning. Under Avg@32—which quantifies the probability of producing a correct response under high-entropy sampling—DCPO exhibits markedly greater robustness at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B attains 38.8 versus GRPO (32.1), DAPO (31.6), and GSPO (34.9); on AIME25-Avg@32, DCPO-14B attains 19.0 versus GRPO (10.5), DAPO (15.3), and GSPO (9.9). These margins indicate that DCPO improves not only expected accuracy but also the diversity and reliability of the correct generations—properties that dominate performance under sampling-based decoding, where temperature-driven variability governs the output distribution. The capacity to jointly optimize reward fidelity and exploration under high-entropy decoding is precisely what is required in deployment settings that draw multiple candidate outputs per query.

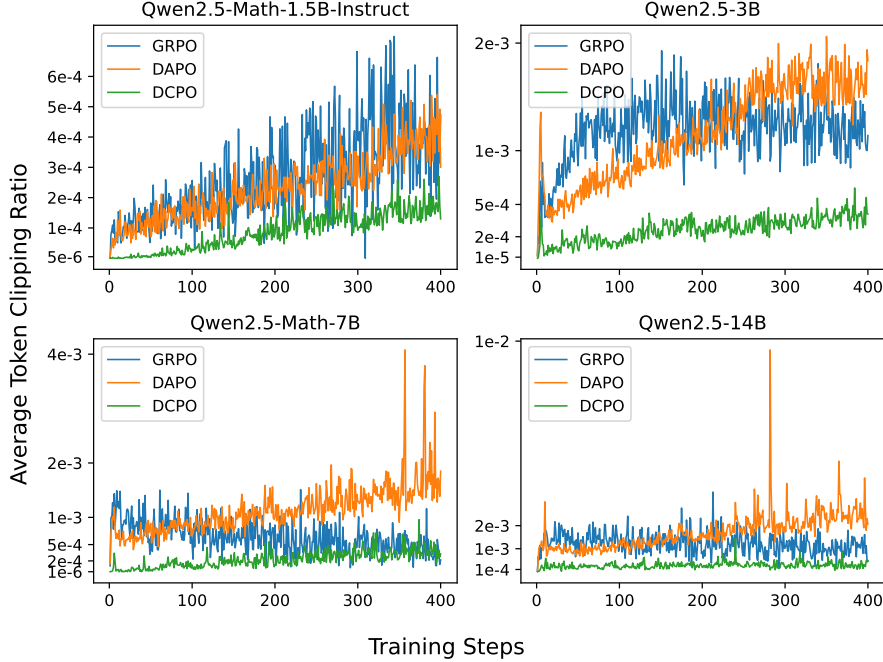


Figure 1: Token Clipping Ratio (TCR) as a function of training step, across models and methods.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio (TCR)** as the fraction of tokens whose gradient is suppressed by the clipping operator during back-propagation, and hence excluded from the policy update. As Figure 1 shows, the trajectory of the TCR over training differs substantially across the reinforcement-learning methods considered.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

Under GRPO, the TCR trajectory is capacity-dependent: for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) it increases monotonically with training steps, whereas for the larger models

(Qwen2.5-Math-7B and Qwen2.5-14B) it gradually decreases. DAPO, by contrast, exhibits a monotonically increasing *TCR* across all models, so that an ever-larger fraction of tokens is clipped as training proceeds and a growing share of updates is computed from partial or truncated responses. Both GRPO and DAPO further display large fluctuations and intermittent spikes in the *TCR*, which can inject instability into the optimization and precipitate entropy collapse. GSPO incurs an average *TCR* exceeding 11% on Qwen2.5-Math-7B and exceeding 15% on Qwen2.5-14B—far above the token-level clipping methods—and therefore discards a larger volume of informative tokens.

DCPO, in contrast, sustains a stable and efficient training regime. Independent of model capacity and training phase, its *TCR* remains approximately constant and an order of magnitude below that of GRPO and DAPO. This lower and more stable *TCR* implies that DCPO clips fewer tokens, so that a larger fraction of complete response sequences contributes to each update and more of the update budget is allocated to exploring rare tokens. The result is higher sample efficiency than the baselines while the range of outlier variation remains tightly bounded.

Consider Qwen2.5-Math-7B as a representative case. The token distribution rapidly concentrates on high-confidence tokens: after roughly 60 steps, about 95% of generated tokens satisfy $q(x) > 0.9$; this fraction rises to about 97% by roughly 100 steps and continues to grow thereafter. As illustrated in Figure 4d, once the old probability satisfies $q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$, the new probability $p(x)$ contributes to the update within the interval $[\frac{1}{1+\epsilon}, 1]$ and is never clipped, regardless of the clipping scheme. Consequently the overwhelming majority of tokens fall in a region that is never clipped, which accounts for the low aggregate *TCR* observed empirically. Meanwhile, Figure 1 shows the *TCR* rising in the later phase of most runs, indicating that a growing fraction of low-probability tokens is discarded and that the model’s capacity to explore rare tokens is correspondingly curtailed.

Prior work (Wang et al., 2025) establishes that **high-entropy tokens**—equivalently, low-probability tokens—are the principal drivers behind the emergence of reasoning capability. Since entropy is a decreasing function of probability, rarer tokens carry more information; granting such tokens a wider admissible update region therefore enlarges the effective exploration support of the policy and facilitates the acquisition of diverse reasoning behaviors.

These empirical and theoretical observations motivate the proposed **dynamic-adaptive clipping** scheme, which relaxes the clipping interval for tokens with $q(x) < \frac{1}{1+\epsilon}$ while retaining the tight interval for high-confidence tokens. By selectively widening the update region over the high-entropy tail of the distribution, the scheme restores the policy’s ability to learn from rare, informative tokens and thereby improves its reasoning performance.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a response contributes to the policy update only if its advantage—and hence its gradient—is nonzero. We quantify this contribution with the **Response Utilization Ratio** (*RUR*), defined as the fraction of generated responses that carry a nonzero advantage and therefore participate in the update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

As Table 2 shows, the average *RUR* under GRPO is low—a mean of only 44.6% across the four models—so that more than half of all generated responses are discarded, evidencing pronounced inefficiency in response utilization. DCPO instead sustains an average *RUR* of approximately 70% after the first epoch, an absolute gain of 28 percentage points and a relative gain of 64% over GRPO.

Table 2: Average *RUR* computed over 400 training steps. Entries marked “-” indicate that GSPO was not evaluated on the corresponding model.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that the *RUR* under GRPO decays substantially over training, independent of model size or type, falling from above 90% to below 50%; for Qwen2.5-Math-7B it drops to as low as 30%

after approximately 200 steps. This decay is a direct consequence of step-local advantage standardization, which produces a growing fraction of zero gradients and renders exploration increasingly inefficient in the later phase of training. DCPO, by employing smooth advantage standardization, instead maintains a consistently high *RUR*: after the first epoch it stabilizes near 70% and improves further with continued training.

We observe that the *RUR* exceeds 90% during the first few steps but, from the second epoch onward, settles near 70% without reaching 100%. We attribute these two regimes to distinct causes and use them to characterize the operating behavior of the SAS mechanism.

First, the initial *RUR* above 90% arises predominantly because the early-stage model produces outputs that violate the required format, which incur negative rewards and thus inflate the *RUR*. Because the models learn the fixed answer format rapidly, later training should shift emphasis toward eliciting reasoning capability rather than formatting compliance. Second, our analysis identifies two mechanisms that yield persistently zero advantages and thus cap the steady-state *RUR* near 70% rather than 100%: (i) roughly half of the problems are sufficiently easy that the model solves them correctly on every sample, and excluding these from the update is justified because the corresponding capabilities require no further refinement; (ii) the remaining zero-advantage instances are those the model answers incorrectly on every sample, which in turn has two sources—some problems exceed the current model’s capability even given a correct ground-truth label, and, separately, the DAPO-Math-17K corpus contains items whose labels were coerced into mislabeled integers, introducing spurious “incorrect” targets. In the latter case, discarding zero-advantage responses functions as an implicit filter that reduces the risk of updating the policy on noisy or mislabeled data.

Because DAPO—unlike GRPO and DCPO—removes prompts whose responses share identical rewards from the generated data, a direct *RUR* comparison is ill-defined. We therefore assess DAPO’s efficiency by the number of generated responses and the wall-clock training time required to reach a fixed number of update steps. To offset the low sampling efficiency induced by a low *RUR*, we set DAPO’s generation batch size to $1\text{--}4\times$ the training batch size and adjust it dynamically during training. Across the four models, DAPO requires $3\text{--}5\times$ more generated responses than DCPO to reach the same number of parameter updates, which at least doubles the training GPU-hours without improving performance. Moreover, DAPO’s utilization degrades with model scale: on Qwen2.5-14B, the fraction of responses used falls below 30% after approximately 300 steps. Thus dynamic sampling attains low data utilization and training efficiency and forgoes exploration of model diversity.

In summary, DCPO advances substantially over both GRPO and DAPO by markedly increasing response utilization and training efficiency, exploiting the sampled responses more fully to widen the exploration support and, in turn, improve task performance.

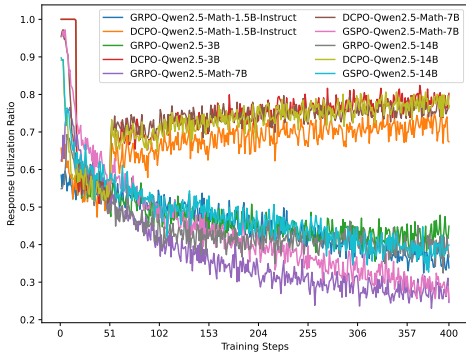


Figure 2: Evolution of the *RUR* over training steps.

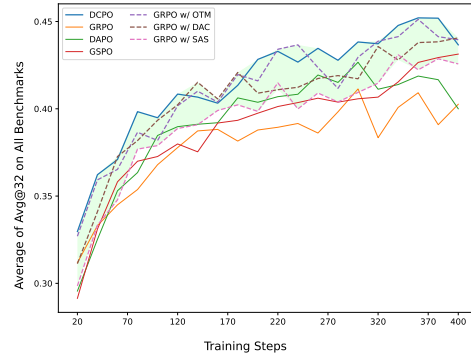


Figure 3: Component ablation on Qwen2.5-Math-7B, measured by average Avg@32 over training steps.

5.4 ABLATION STUDIES

We conduct a component ablation on Qwen2.5-Math-7B to isolate the contribution of each element of DCPO, using Avg@32 as the metric because it reflects the robustness and stability of the learned policy distribution. For a controlled comparison, each variant modifies exactly one component of the GRPO baseline while holding all other settings fixed and removing the KL-divergence term, so as

to match DAPO, GSPO, and the full DCPO. The learning curves over steps 20–400 are reported in Figure 3.

Each of the three single-component variants improves reasoning accuracy monotonically over training, exceeding GRPO and matching or surpassing DAPO and GSPO. Thus each modification confers a distinct benefit, and their combination yields a large aggregate gain over the baselines, establishing the effectiveness of the DCPO pipeline. In the later training phase the baselines plateau, whereas DCPO continues to improve at both the component and full-pipeline level. The ablation confirms that each component contributes positively and that their integration produces cumulative gains and improved stability; the per-component effects are detailed below.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within each response rather than across the batch yields a model that outperforms the baselines, which suffer from lower data utilization and larger performance fluctuations. (ii) **Smoothed Advantage Standardization (SAS)**: replacing the step-local advantage with the cumulative smoothed advantage substantially improves over GRPO and approaches DAPO, demonstrating that aggregating reward statistics across steps stabilizes optimization and raises final accuracy. (iii) **Dynamic Adaptive Clipping (DAC)**: replacing GRPO’s fixed clipping interval with the probability-dependent adaptive interval yields the largest single-component gain—DAC alone exceeds GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **OTM + SAS + DAC (full DCPO)**: combining the three mechanisms attains the best and most stable results, significantly exceeding GRPO, DAPO, and GSPO. Their synergy indicates that each design choice contributes complementarily to data-efficient and stable reinforcement learning, improving both accuracy and robustness.

Overall, the ablation confirms that every component of DCPO contributes positively and that their combination produces substantial cumulative gains, validating the proposed mechanisms as effective means of improving data efficiency and stability in reinforcement learning for LLMs.

6 CONCLUSION

This work addresses two structural limitations of existing policy-optimization methods: restricted token-level exploration caused by a fixed clipping interval in low-probability regions, and low sample efficiency caused by step-local reward standardization. We introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement-learning pipeline that improves LLM reasoning through two mechanisms: (i) a **dynamically adaptive token-level clipping** operator whose bounds are an explicit function of the old-policy probability, enabling more efficient exploration of low-probability tokens, and (ii) a **cumulative smooth advantage standardization** that aggregates reward statistics across optimization steps, suppressing the reward variance introduced by high-entropy sampling. Experiments on four mathematical-reasoning benchmarks across four models establish the effectiveness of DCPO: the DCPO-7B variant attains an Avg@32 of 38.8 on AIME24, exceeding GRPO (32.1) and DAPO (31.6); DCPO further raises the response utilization ratio by 28 percentage points over GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude. These results demonstrate that DCPO substantially improves data utilization and exploration in RLVR. Future work will extend DCPO to additional domains, including code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In prior methods, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ of token t in response j is obtained by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards of the G responses generated at step i , as given in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for a prompt are identical across the group, the standardized advantage is zero and the corresponding responses contribute no gradient, so the generated responses are wasted.

A.3.1 GRPO

Shao et al. (2024) introduced **Group Relative Policy Optimization (GRPO)**, which samples G responses per query, assigns rewards R via a rule-based reward function, estimates token-level advantages as in Equation (11), and updates the parameters with Equation (12), termed the **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages the token-level advantages over the tokens of each response and then over the G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty constrains the policy against a reference policy π_{ref} (e.g., the pre-trained model), and the fixed threshold ε bounds the per-token probability-ratio change to prevent overly large updates. This nested averaging, though widely adopted, dilutes the relative importance of individual responses.

A.3.2 DAPO

Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, observing that in GRPO longer sequences exert disproportionate influence on the gradient. DAPO normalizes by the total token count over the batch, yielding the **Token Level Mean loss (TLM)** of Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. DAPO discards any prompt whose G responses share an identical reward and regenerates responses to preserve the batch size, which incurs severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping injects high-variance noise into the gradient estimate, that this noise accumulates over long sequences, and that the clipping operator further amplifies it. GSPO replaces token-wise clipping with sequence-wise clipping to obtain a more stable and effective training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

where $s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)}\right)\right)$

GSPO discards more than 10% of the responses that carry nonzero advantages, incurring substantial data waste and reduced training efficiency. Conversely, while GRPO suppresses responses exhibiting high sequence-level variance, it retains many responses with high token-level entropy, which destabilizes training.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we adopt a rule-based reward function that scores each response jointly on answer correctness and format compliance, as defined in Equation (15). The function assigns +1 when both the format and the answer are correct, 0 when the format is correct but the answer is incorrect, and -1 when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response within the group for a given problem at training step i , and the reward is determined by the format validity and answer correctness of that response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a key mechanism for stabilizing policy optimization and bounding the per-step parameter displacement. It is especially important under importance sampling, where expectations under a new policy are estimated from samples drawn by an old policy.

The expectation of a function $f(x)$ under the new probability $p(x)$ can be expressed as an expectation under the old probability $q(x)$ via the importance weight $p(x)/q(x)$, as in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although unbiased, this estimator can exhibit substantially inflated variance—a well-known pathology of importance sampling. The on-policy variance of $f(x)$ under the new policy is given by Equation (17), while the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The gap between the importance-sampled variance and the on-policy variance quantifies the inflation, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To control this gap, many prior methods—including PPO Schulman et al. (2017) and RLVR variants (DeepSeek-AI et al., 2024; Liu et al., 2025)—impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently limits the policy’s exploration capacity, which is especially problematic in regions where the old policy places low probability mass: there the admissible absolute change is tightly constrained, restricting the magnitude of any meaningful update. Asymmetric bounds, as in DAPO, mitigate this only partially, since the non-adaptivity of the constraint persists.

Furthermore, Yue et al. (2025) report that the reasoning paths exploited by RLVR models lie largely within the base model’s sampling distribution, a claim corroborated by the small parameter displacement observed during off-policy batch updates and reflected in the small fraction of clipped parameters. Motivated by this stability, we constrain the probability ratio $r(x)$ through a dynamic-adaptive mechanism that incorporates the probability itself into the constraint, as in Equation (21), replacing the fixed clipping of Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For the resulting clipping interval to be mathematically well-defined, the constraint must respect the elementary properties of probability distributions and their ratios. As stated in Equation (23), the probabilities $p(x)$ and $q(x)$ lie in $[0, 1]$; consequently the ratio $r(x)$ is non-negative, and the clipping hyperparameter ϵ is likewise non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure the validity and practical applicability of the interval derived from Equation (22), we state the required conditions on its bounds in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24), the probability ratio $r(x)$ must satisfy the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

To define the clipping interval, we take the overlapping region centered at the ratio value 1, which minimizes the discrepancy between the two conditions and preserves consistency with the conventional fixed bound of GRPO. This yields the following validity conditions on the interval:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, to keep the radicand non-negative, we clamp it against zero via $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This yields the following conditions, which constitute the **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

In Inequality 27, we use distinct upper and lower thresholds ϵ_{high} and ϵ_{low} so that the adaptive interval coincides with the GRPO interval at prescribed anchor points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To make the dynamic-adaptive bounds coincide with the fixed symmetric bounds at the anchor points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we solve for ϵ_{low} and ϵ_{high} from the system in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

With the fixed threshold ϵ set to its customary value 0.2 (as in GRPO), solving the system gives approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Here the inequality $\epsilon_{low} \neq \epsilon_{high}$ serves a purpose distinct from the asymmetric fixed clipping of DAPO: it anchors the adaptive interval to the fixed one rather than merely widening the upper bound.

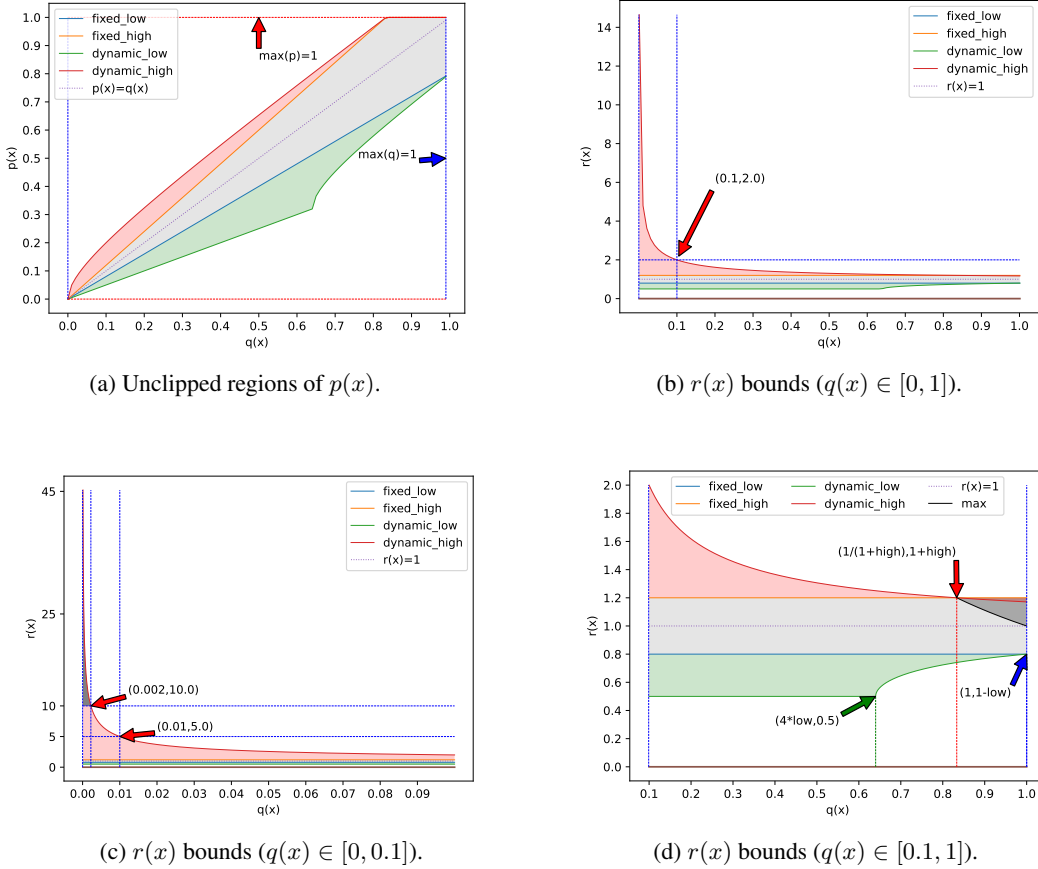


Figure 4: Comparison of clipping bounds. Curves show the admissible interval under fixed clipping ($\epsilon = 0.2$) and under dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$), as functions of the old probability $q(x)$.

Dynamic versus fixed clipping: Figure 4 contrasts the interval under GRPO’s fixed clipping ($[1 - \epsilon, 1 + \epsilon]$) with that under our probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed scheme imposes a constant relative width 2ϵ , so its admissible *absolute* change shrinks linearly as $q(x)$ decreases (Figure 4a). By contrast, the relative width of the adaptive scheme scales inversely with $q(x)$, keeping the effective exploration region approximately constant across the probability range.

Effect of decreasing $q(x)$: Because the fixed interval $[1 - \epsilon, 1 + \epsilon]$ is independent of $q(x)$, it does not enlarge the exploration region for low-probability tokens as $q(x)$ decreases; the adaptive bound, in contrast, expands rapidly (Figure 4b), granting rare tokens a proportionally larger admissible region—precisely the behavior required for efficient RL.

Behavior for extremely low probabilities: Figure 4c plots the interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the adaptive upper bound on $r(x)$ diverges at a rate inversely proportional to $\sqrt{q(x)}$, whereas the fixed bound remains constant. Following the dual-clipping strategy of (Ye et al., 2020), we impose a hard ceiling $r_{\max} = 10$ to preclude the attendant instabilities (vanishing or exploding gradients). The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper bound is saturated at $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$ the adaptive scheme yields $dynamic_{low} = 0.5$, whereas fixed clipping retains the constant $fixed_{low} = 1 - \epsilon = 0.8$; as $q(x)$ exceeds this threshold, the adaptive lower bound contracts smoothly toward the fixed value. Symmetrically, the adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\epsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ therefore widens the admissible region for rare tokens in a probability-dependent manner while preserving the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The objective in Equation (8) comprises both a positive-advantage and a negative-advantage term. Under the minimum operation, the ratio $r(x)$ for a positive advantage is clipped to $[0, dynamic_{high}(q(x), \epsilon_{high})]$, and for a negative advantage to $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. To prevent an excessively large $r(x)$ from destabilizing training, we cap the ratio at $r_{\max} = 10$ for both advantage signs. This safeguard, following Ye et al. (2020), ensures the adaptive bounds do not induce excessively large importance weights.

In summary, for both advantage signs, dynamic-adaptive clipping affords the model a larger exploration region than fixed clipping when rare tokens have low probability.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i index the iteration at which prompts are used to generate responses and update the parameters. We define the following reward-normalization statistics.

The reward means are given by the family of equations in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

The reward variances are given by the family of equations in Equation (30):

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce computational and memory cost, we evaluate σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29) through the equivalent recurrences in terms of μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i , rather than recomputing them directly from the raw rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments employ the following prompt template, adapted from the Qwen-Math template (Yang et al., 2024b).

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

We configure the clipping thresholds per method as follows: a symmetric threshold $\epsilon = 0.2$ for GRPO; asymmetric lower/upper thresholds (0.2, 0.28) for DAPO, together with a 512-token soft-punishment cache and a maximum probability ratio of 10 for negative advantages; lower/upper thresholds $(3e-4, 4e-4)$ for GSPO; and low/high thresholds (0.16, 0.2) for DCPO. The DCPO thresholds are chosen to preserve the same anchor points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d); we further cap the probability ratio at 10 for *both* positive and negative advantages, in contrast to DAPO’s single-sided cap.

Each run comprises 400 optimization steps with a generation batch size of 512 and a parameter-update mini-batch size of 32. We sample $G=16$ responses per problem at temperature 1 and top_p 1, with a maximum prompt length of 1,024 tokens and a maximum response length of 3,072 tokens—substantially below the 20k limit used in the original DAPO.

Our training codebase is adapted from Verl (Sheng et al., 2024) and runs on 32×H20 GPUs. Unless noted otherwise, hyperparameters match those of DAPO. For the loss, we retain the exact $\frac{1}{G}$ (GRPO) or $\sum_{i=1}^G \frac{1}{|\alpha_i|}$ (DAPO) weighted-average formulation rather than Verl’s approximate weighting, ensuring correctness across differing microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports the mean Avg@1 metric, defined as the average of the greedy-decoding (Avg@1) scores over MATH500, AMC23, AIME24, and AIME25. DCPO consistently exceeds GRPO and DAPO on all four models, and exceeds GSPO on the two models where GSPO was evaluated. Its trajectory is also more stable than the baselines’, exhibiting neither the pronounced performance drop of GRPO nor the large fluctuations of DAPO. This indicates that DCPO elicits stronger reasoning capability during training and affords greater headroom for further performance gains.

Avg@32 Trend Figure 6 reports the Avg@32 trajectories of DCPO against GRPO, DAPO, and—where available—GSPO across model sizes on AMC23, AIME24, and AIME25, and demonstrates the robustness and stability of DCPO. GRPO, by contrast, exhibits large fluctuations and occasional collapse; on Qwen2.5-14B it suffers a marked drop after approximately 170 steps. Across all models (1.5B to 14B), the DCPO trajectory is monotonically increasing with minimal fluctuation, attains the fastest early-stage growth rate, and sustains its upward trend. Relative to DAPO, DCPO attains superior performance while halving training time and resource consumption. These results establish the effectiveness and robustness of dynamic-adaptive clipping and smooth advantage standardization for improving both response-level and label-level exploration diversity.



Figure 5: Avg@1 performance over training steps, across benchmarks.

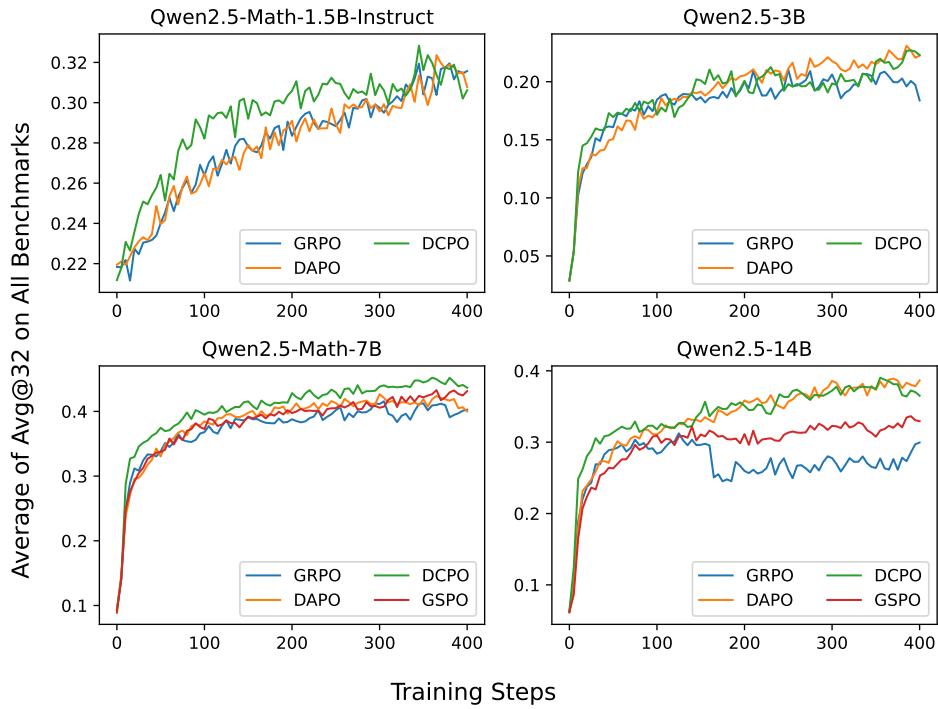


Figure 6: Avg@32 performance over training steps, across benchmarks.

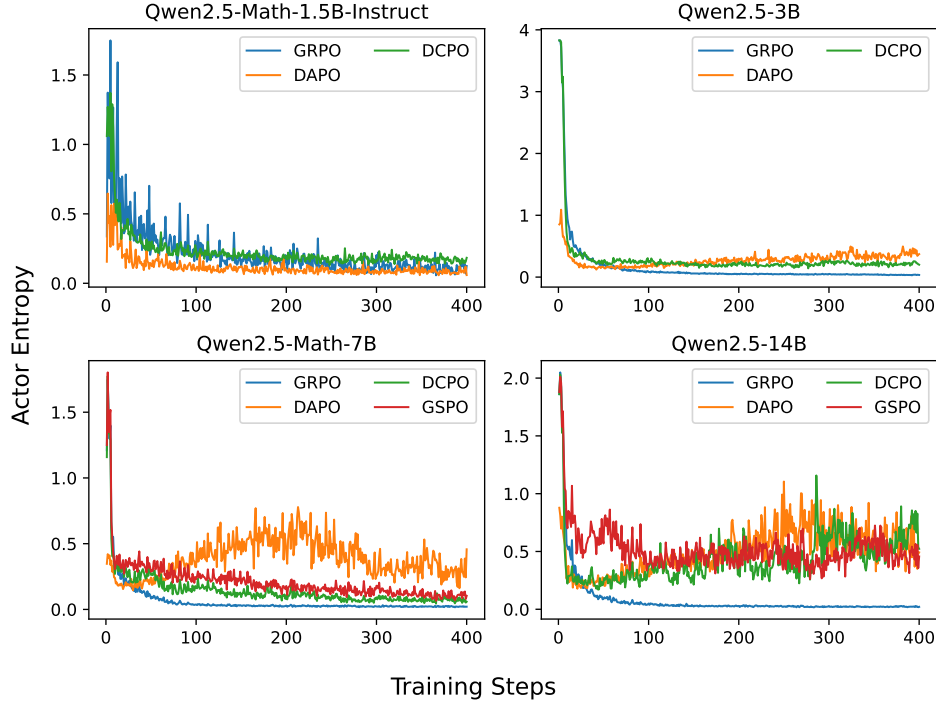


Figure 7: Policy entropy over training steps, across models and methods.

Entropy Trend Figure 7 shows that GRPO undergoes a sharp entropy decay after an initial warm-up phase and then plateaus in a flat, low-entropy regime. This entropy collapse corresponds to a rapid loss of policy stochasticity, which reduces the diversity of sampled responses and constrains subsequent optimization.

The behavior of DAPO and DCPO is capacity-dependent. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow comparable declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), however, the trajectories diverge markedly: DAPO sustains the highest entropy, with a clear upward trend and large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy is comparable to DCPO’s, as is its average metric; on Qwen2.5-14B, by contrast, GSPO exhibits a fluctuating downward trend and correspondingly poor performance. DCPO instead settles at a moderate entropy level, with a smoother curve confined to a bounded band between GRPO’s low-entropy regime and DAPO’s high-variance regime.

These observations are consistent with a U-shaped relationship between entropy and performance: excessively low entropy starves the learner of exploratory signal and induces premature convergence, whereas excessively high or unstable entropy degrades training stability. By holding entropy within a moderate, well-regulated band, DCPO promotes both convergence and sufficient exploration—an effect that strengthens with model capacity. This entropy-regulation property is a plausible mechanism underlying DCPO’s superior reasoning performance and its robust policy learning across model scales.